04. AUDITORNE VJEŽBE:

~ Carnot, Brayton-Joule, Rankin ~

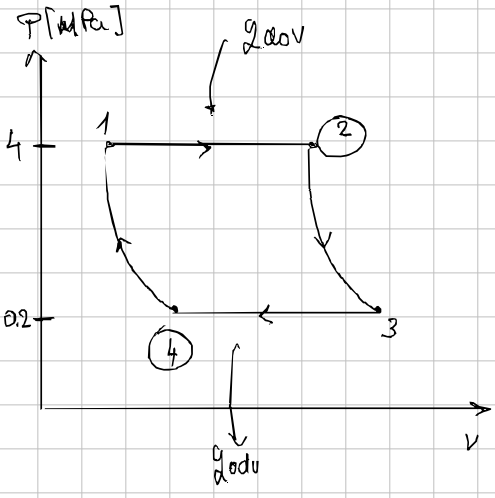
1 U termoelektrani s plinskom turbinom odvija se idealni desnokretni BJ KP. Tlak u komori izgaranja je 4 MPa, a u hadnjaku 0.2MPa.Temperatura plina na izlazu iz komore izgaranja je 1400K, a na izlazu iz hladnjaka je 400K. Pretpostaviti da se proces odvija s idealnim plinom. Plinska konstanta je 287 J/kgK,a adijabatski indeks k = 1.4. Izracunajte:

- temp i specificne volumene plina u 4 karakt tocke procesa
- qdov, qodv, wt dobiveni i wk ulozeni
- termicki stupanje djelovanja

$pV = RT$

1-2 = komora izgaranja

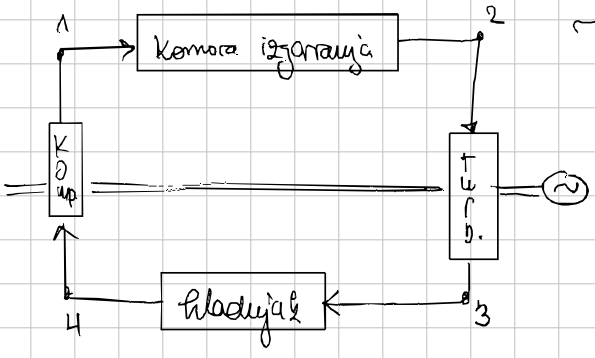
3-4 = hladnjak



$p_1 = p_2 = 4\,MPa$ $\quad p_3 = p_4 = 0.2\,MPa$

$T_2 = 1400K$ $\quad T_4 = 400K$

$R = 287\,J/kgK$ $\quad K = 1.4$



1-2 = IZOBARNO $\quad n = 0$

$$\frac{T_1}{v_1} = \frac{T_2}{v_2}$$

$$q_{12} = c_p (T_2 - T_1) \rightarrow q_{12} = q_{dov} = 460.6\,kJ/kg$$

$$\boxed{v_1 = 0.067\,m^3/kg}$$

2-3 = ADIJABAT $\quad K = 1.4 \quad q_{23} = 0$

$$\frac{T_2}{T_3} = \left(\frac{p_2}{p_3}\right)^{\frac{0.4}{1.4}} \longrightarrow \boxed{T_3 = 594.8\,K}$$

$$p_2 v_2 = RT_2 \quad \boxed{\longrightarrow v_2 = 0.1\,m^3/kg}$$

$$w_t = -c_p \Delta T \qquad c_p = 1004.5 \longrightarrow w_t = -1004.5 \cdot (594.8 - 1400)\,K$$

$$\boxed{w_t = 808.82\,kJ/kg}$$

3-4 = IZOBARNO

$$\frac{T_3}{v_3} = \frac{T_4}{v_4}$$

$$\boxed{q_{odv} = -195.7\,kJ/kg}$$

$$\boxed{\begin{aligned} v_3 &= 0.85\,m^3/kg \\ v_4 &= 0.574\,m^3/kg \end{aligned}}$$

4-1 = ADIJABAT

$$\frac{T_4}{T_1} = \left(\frac{p_4}{p_1}\right)^{\frac{0.4}{1.4}} \longrightarrow \boxed{T_1 = 941.4\,K}$$

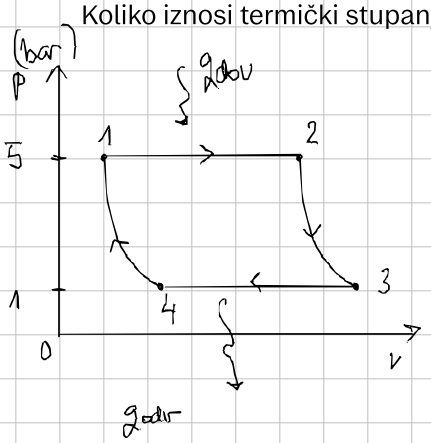
$$\boxed{w_k = -543.8\,kJ/kg}$$

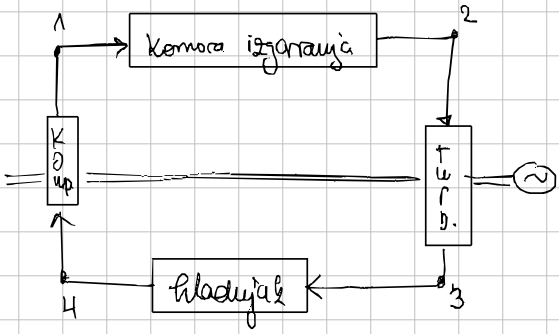
$$\boxed{\eta_t = \frac{w_t - |w_k|}{q_{dov}} = \dots = 0.575}$$

Zadatak 2.)

Desnokretni Braytonov kružni proces s idealnim plinom ($\kappa = 1.4$ i $R = 287$ J/kgK) sastoji se od realne adijabatske kompresije i ekspanzije te izobarnog dovođenja i odvođenja topline. Na ulazu u kompresor plin ima temperaturu 15 °C i tlak 1 bar, a na izlazu iz kompresora tlak 5 bara. Temperatura plina na ulazu u turbinu iznosi 780 °C. Unutrašnji stupanj djelovanja kompresora iznosi 0,83, a turbine 0,85.

Koliko iznosi termički stupanj djelovanja Braytonovog kružnog procesa?





$\vartheta_4 = 15°C \rightarrow T_4 = 288{,}15K$

$p_4 = 100kPa = p_3$

$p_1 = p_2 = 500kPa$

$\vartheta_2 = 780° \rightarrow T_2 = 1063{,}15K$

$\eta_{kompr} = 0{,}83$

$\eta_{turb} = 0{,}85$

$\eta_{t\,B} = ?$ $\quad \dfrac{w_t - |w_k|}{q_{dov}}$

$q_{dov} = c_p(T_2 - T_1)$

4-1 = ADIJABAT

$\dfrac{T_4}{T_1} = \left(\dfrac{p_4}{p_1}\right)^{\frac{0.4}{1.4}} = \left(\dfrac{1}{5}\right)^{\frac{0.4}{1.4}} \rightarrow T_1 = 456.4K$

$\boxed{q_{dov} = 609{,}5\,kJ/kg}$

$w_{t\,turbina} = -c_p(T_3 - T_2)$

2-3 = ADIJABAT $\rightarrow T_3 = 671.3K$

$w_{teor} = 393.6\,kJ/kg$

$\Rightarrow w_t = w_{teor} \cdot 0{,}85$

$\rightarrow \boxed{w_t = 334.6\,kJ/kg}$

$w_{komp} = -c_p(T_1 - T_4) \rightarrow w_{komp} = -169\,kJ/kg$

$w_k = w_{komp} \cdot 0{,}83 \rightarrow \boxed{w_k = -143.7\,kJ/kg}$

$\eta_t = \dfrac{w_t - |w_k|}{q_{dov}} = \dfrac{334.6 - 143.7}{609.5} \Rightarrow \boxed{\eta_t = 0.313}$

Zadatak 2.)

Desnokretni Braytonov kružni proces s idealnim plinom ($\boxed{\kappa = 1.4 \text{ i } R = 287 \text{ J/kgK}}$) sastoji se od realne adijabatske kompresije i ekspanzije te izobarnog dovođenja i odvođenja topline. Na ulazu u kompresor plin ima temperaturu 15 °C i tlak 1 bar, a na izlazu iz kompresora tlak 5 bara. Temperatura plina na ulazu u turbinu iznosi 780 °C. Unutrašnji stupanj djelovanja kompresora iznosi 0,83, a turbine 0,85.

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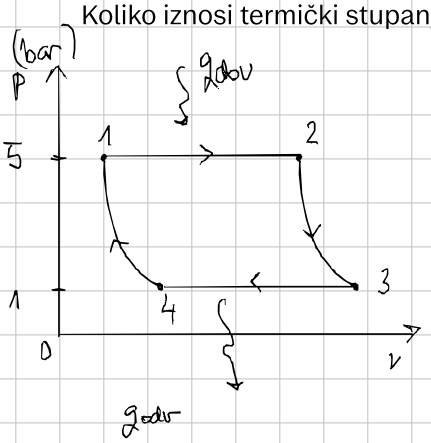

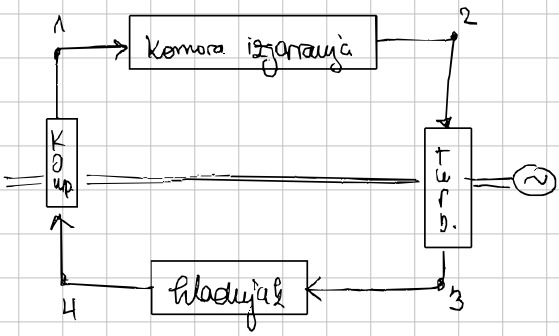

$\vartheta_4 = 15°C \longrightarrow T_4 = 288{,}15K$

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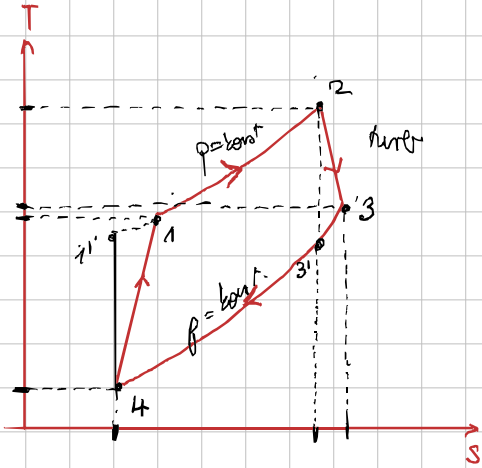
$\eta_{kompr} = 0{,}83$

$\eta_{turb} = 0{,}85$

$\eta_{t\,B} = ?$



UNUTRAŠNJI ST. DJELOVANJA: $\eta_{ut} = \dfrac{w_{real\,t}}{w_{ideal\,t}}$

$$\eta_{ut} = \frac{T_3 - T_2}{T_3' - T_2}$$

$T_3' \longrightarrow$ ADIJABATA : $\dfrac{T_3'}{T_2} = \left(\dfrac{p_3'}{p_2}\right)^{\frac{0.4}{1.4}} \ldots \longrightarrow T_3' = 664{,}9K \longrightarrow T_3 = 723.1K$

$\eta_{unut\,K} = \left(\dfrac{ideal}{real}\right)^{*} = \dfrac{T_1' - T_4}{T_1 - T_4} \ldots \longrightarrow T_1' = 456.4K \longrightarrow T_1 = 490.9K$

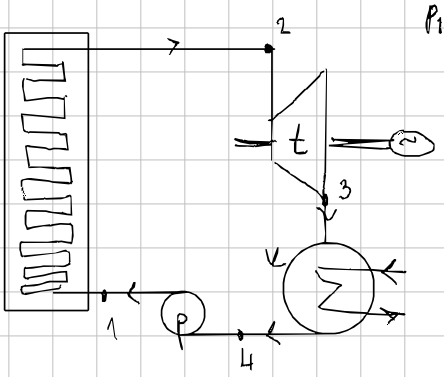
$$\eta_t = \frac{w_t - |w_k|}{q_{dov}} = \frac{\cancel{c_p}(T_2 - T_3) + \cancel{c_p}(T_4 - T_1)}{\cancel{c_p}(T_2 - T_1)} = \frac{T_2 - T_1 + T_4 - T_3}{T_2 - T_1} = 1 + \frac{T_4 - T_3}{T_2 - T_1}$$

$$\boxed{\eta_t = 0{,}226}$$

*idealno: $1 - \dfrac{T_4}{T_1'} = 0{,}369$

Zadatak 3.)

U termoelektrani s parnom turbinom odvija se realni Rankineov kružni proces. Na ulazu u turbinu tlak je pare 10 MPa, temperatura 700 °C, entalpija 3867 kJ/kg, a entropija 7.166 kJ/kgK. Tlak je u kondenzatoru 10 kPa. Entalpija mokre pare na tlaku 10 kPa i entropiji 7.166 kJ/kgK je 2271 kJ/kg. Entalpija vode na izlazu iz kondenzatora je 192 kJ/kg. Unutrašnji stupnjevi djelovanja turbine i pumpe iznose 0.87. Koliko iznosi termički stupanj djelovanja kružnog procesa? Računajte s konstantnim specifičnim volumenom kondenzata (vode) što ga pojna pumpa vraća u kotao, v = 0.001 m3/kg.



$p_1 = p_2 = 10\,MPa$ $\qquad$ $p_3 = p_4 = 10\,kPa$

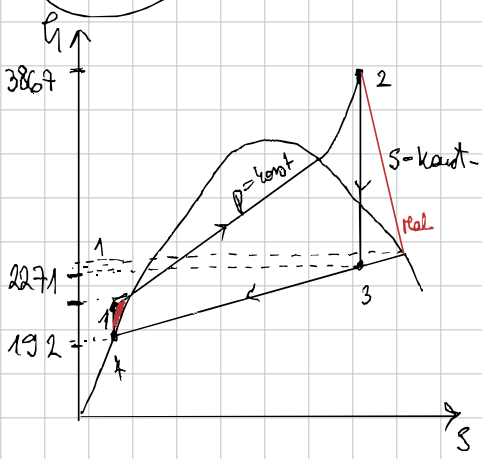
$\vartheta_2 = 700\,°C$ $\qquad$ $s_3 = 7.166\,kJ/kgK$

$h_2 = 3867\,kJ/kg$ $\qquad$ $h_3 = 2271\,kJ/kg$

$s_2 = 7.166\,kJ/kgK$ $\qquad$ $h_4 = 192\,kJ/kg$

$\eta_u = 0.87$ (turbina i pumpa)

$\eta_{tR} = ?$ $\qquad$ $v = 0{,}001\,m^3/kg$



$h_1 = h_4 + w_{t\,pump}$

$q_{23} = h_2 - h_1$ $\qquad$ $w_t = h_2 - h_3 \rightarrow w_t = 1596\,kJ/kg$

$q_{34} = h_4 - h_3$ $\qquad$ $w_p = h_4 - h_1$ $\qquad$ $\boxed{w_t = 1388.5}$

$w_p = -v\int_{p_4}^{p_1} dp \rightarrow w_p = -9{,}99\,kJ/kg$

tehnički rad $= -w_p \rightarrow w_{teh\,p} = 9{,}99\,kJ/kg$

$\eta_u = \frac{w_{ideal}}{w_{real}} \rightarrow w_{real\,p} = \frac{w_{id\,p}}{\eta_u} \rightarrow \boxed{w_{real\,p} = 11.483\,kJ/kg}$

tehnički

$q_{dov} = h_2 - h_1$? $\qquad$ $h_1 = h_4 + w_{real\,(teh)\,pumpa} = 192 + 11.483$

$h_1 = 203.483\,kJ/kg \rightarrow \boxed{q_{dov} = 3663.52\,kJ/kg}$

$\eta_{tR\,(real)} = \frac{w_t - |w_p|}{q_{dov}} \rightarrow \boxed{\eta_{tR\,(real)} = 0.376}$

→ Kada bi se radilo o idealnom: $\eta_{tid} = \frac{1596 - 9{,}99}{3867 - 192 - 9{,}99} \rightarrow \boxed{\eta_{id\,t} = 0{,}433}$

Zadatak 4.)

Snaga parne turbine u termoelektrani je 50 MW. Tlak je pare na ulazu u turbinu (unutrašnjeg stupnja djelovanja 0,85) 7 MPa, a temperatura 500 °C. Unutrašnji je stupanj djelovanja pumpe 0,85. Tlak je u kondenzatoru 20 kPa, a održava se pomoću rashladne vode masenog protoka 5000 kg/s. Specifični je toplinski kapacitet rashladne vode 4180 J/kgK. Iz parnih tablica očitane su sljedeće karakteristične vrijednosti: za tlak 20 kPa: h' = 251,5 kJ/kg, h'' = 2610 kJ/kg, s' = 0,832 kJ/kgK, s'' = 7,909 kJ/kgK, v' = 0,001 m3 /kg; za tlak 7 MPa i temperaturu 500 °C: h = 3411 kJ/kg, s = 6,799 kJ/kgK. Odredite termički stupanj djelovanja Rankineovog kružnog procesa i porast temperature rashladne vode.

$P_t = 50\,MW$

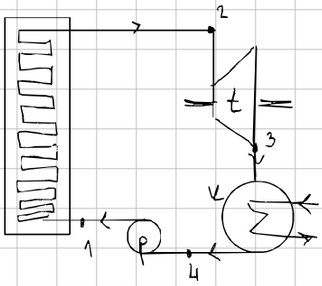
$p_2 = p_1 = 7\,MPa$

$\eta_{ut} = 0{,}85$

$p_3 = p_4 = 20\,kPa$

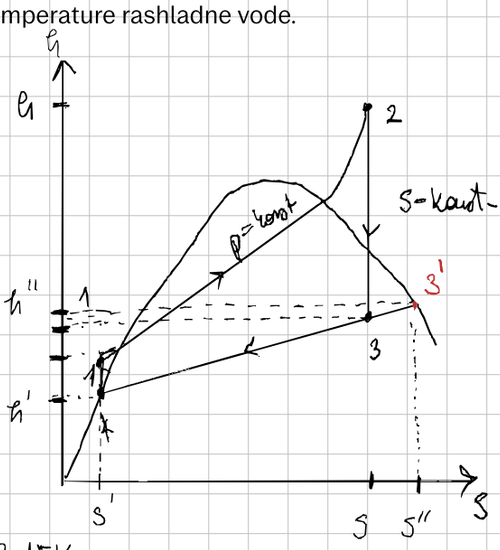
$\dot m = 5000\,kg/s$

$c_v = 4180\,J/kgK$



$\eta_{tR} = ?$

$v = 0{,}001\,m^3$



	20 kPa
h'	251.5 kJ/kg
h''	2610 kJ/kg
s'	0,832 kJ/kgK
s''	7,909 kJ/kgK

	7 MPa, T = 773,15 K
h	3411 kJ/kg
s	6,799 kJ/kgK

$q_{12} = h_2 - h_1$

$q_{34} = h_4 - h_1$

$w_t = h_2 - h_3$ ili $w_t = v\int_{p_2}^{p_3} dp$

$w_p = h_4 - h_1$ ili $w_p = -v\int_{p_4}^{p_1} dp$

$h_3 = h' + x(h'' - h')$

$x = \frac{s - s'}{s'' - s'} \rightarrow x = 0.84 \rightarrow h_3 = 2232.64\,kJ/kg$

$w_{t,id} = 3411 - 2232.64 \Rightarrow w_{t,id} = 1001.86\,kJ/kg$ (real)

$\eta_{ut} = \frac{w_{real}}{w_{id}}$ (jer smo već izračunali ih) $\Rightarrow w_{real(t)} = 1001.86\,kJ/kg$

$\eta_{it} = \frac{h_2 - h_3}{h_2 - h_3'}$

$h_3 = 2409.14$

$w_{teh\,pumpa} = h_1 - h_4 \Rightarrow w_{teh\,pumpa} = v\int_{p_4}^{p_1} dp = 0{,}001(7\,MPa - 20\,kPa)$

$w_{teh\,pumpa} = 6.98\,kJ/kg$ (idealan)

$\eta_{up} = \frac{ideal}{real}$

$w_{teh\,(real\,pumpa)} = \frac{w_{teh\,id\,p}}{\eta_{up}} = \frac{6.98\,kJ/kg}{0.85} \Rightarrow w_{t\,real} = 8.21\,kJ/kg$

$q_{dov} = h_2 - h_1 \rightarrow h_1 = h_4 + w_{teh\,pumpa\,(real)} = h' + w_{t\,real}$

$q_{dov} = 259.71\,kJ/kg$

$$\boxed{\eta_{t\,real} = \frac{1001.6 - 8.21}{\underbrace{3411 - 259.71}_{q_{dov}}} = 0.315}$$

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34 → rashladna voda

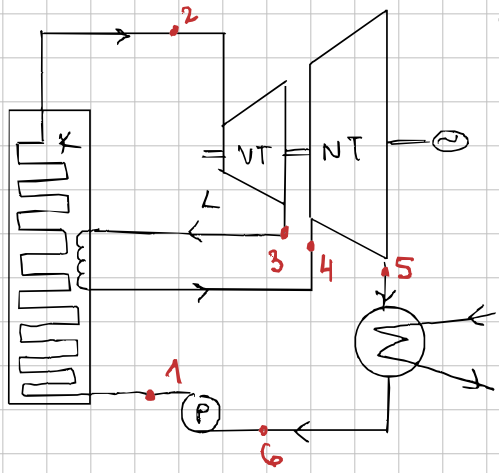
$Q_{34} = m_v c_v \cdot \Delta T = \dot m(h_3 - h_4) = 50(2409.14 - 251.5) = 107.88 \times 10^3$

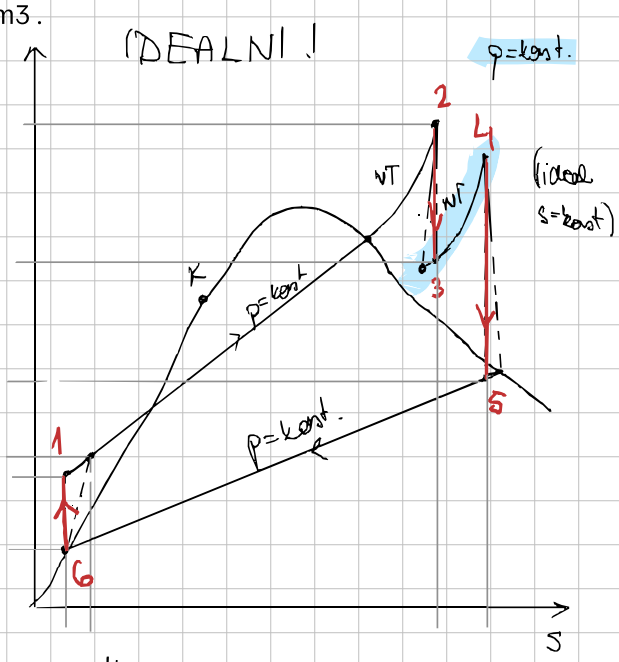
$\dot m = \frac{P_t}{w_t} = \frac{50 \cdot 10^6}{1001.86 \times 10^3} \rightarrow \dot m = 50\,kg/s$ $\quad * \dot m \neq \dot m_v$

$\rightarrow \boxed{\Delta T = \ldots = 5.16\,K}$

Zadatak 5.)

U termoelektrani se odvija idealni Rankineov kružni proces. Para tlaka 7 MPa i temperature 500 °C napušta kotao i ulazi u visokotlačni (VT) dio turbine gdje ekspandira do tlaka 0.9 MPa. Para se zatim dodatno zagrijava do temperature 450 °C te ekspandira u niskotlačnom (NT) dijelu turbine do tlaka 10 kPa. Iz parnih su tablica očitanje vrijednosti entalpija: na izlazu iz kotla 3411 kJ/kg, na izlazu iz VT dijela turbine 2856 kJ/kg, na izlazu iz međupregrijača 3372 kJ/kg, na izlazu iz NT dijela turbine 2431 kJ/kg i na izlazu iz kondenzatora 192 kJ/kg. Izračunati termički stupanj djelovanja kružnog procesa. Gustoća vode što ju pumpa ubrizgava u kotao je 1000 kg/m3 .





$p_1 = p_2 = 7\,MPa$

$T_2 = 773.15\,K$ $\quad h_2 = 3411\,kJ/kg$

$p_3 = 0{,}9\,MPa = p_4$ $\quad h_3 = 2856\,kJ/kg$

$T_3 = 1223.15\,K$ $\quad h_4 = 3372\,kJ/kg$

$p_5 = 10\,kPa = p_6$ $\quad h_5 = 2431\,kJ/kg$

$\rho = 1000\,kg/m^3$ $\quad h_6 = 192\,kJ/kg$

$\eta_t = ?$

$$\eta_t = \frac{W_{tVT} + W_{tNT} - |W_p|}{q_{dov_1} + q_{dov_2}}$$

$q_{dov_1} = q_{12}$ (kotao)

$q_{dov_2} = q_{34}$ (međupregrijač)

$q_{dov_1} = h_2 - h_1$

$W_{tVT} = h_2 - h_3$

$q_{dov_2} = h_4 - h_3$

$W_{tNT} = h_4 - h_5$

$W_p = h_6 - h_1 = -v\int_{p_6}^{p_1} dp$

$h_1 = h_6 + W_{tp} \longrightarrow W_{tp} = -W_p$, $v\int_{p_6}^{p_1} dp$

$$\eta_t = \frac{h_2 - h_3 + h_4 - h_5 - v\int_{p_6}^{p_1} dp}{h_2 - h_6 - v\int_{p_6}^{p_1} dp + h_4 - h_3} = \dots =$$

$$\boxed{\eta_t = 0{,}399}$$

PAZI MJERNE JEDINICE!

* ako nema međupregrijača:

$$\eta_t = \frac{h_2 - h_5 - |W_p|}{h_2 - h_6 + W_p} = 0{,}303$$

znači, poveća se η_t uvođenjem međupregrijača!

Zadatak 6.)

6. Voda protoka 500 kg/s ulazi u kotao na tlaku 10 MPa i temperaturi 30 °C (h = 134.7 kJ/kg, v = 0.001 m3 /kg) u cjevovod promjera 30 cm. Para napušta kotao s tlakom 10 MPa i temperaturom 400 °C (h = 3100 kJ/kg, v = 0.0264 m3 /kg). Odredite toplinsku snagu kotla. Zanemariti promjenu potencijalne energije fluida u kotlu.

OTVORENI SUSTAV:

$\dot{m} = 500\,kg/s$

$E_p = \emptyset$

$$q_{12} + h_1 + \frac{c_1^2}{2} = \cancel{w_{12}} + h_2 + \frac{c_2^2}{2}$$

$w_{12} = \emptyset$

$p_1 = p_2 = 10\,MPa$

$$q_{12} + h_1 + \frac{c_1^2}{2} = h_2 + \frac{c_2^2}{2}$$

$T_1 = 303.15\,K$ $\quad T_2 = 673.15\,K$

$h_1 = 134.7\,kJ/kg$ $\quad h_2 = 3100\,kJ/kg$

$\dot{m}_v$ (voda) $= \dot{m}_p$ (para)

$v_1 = 0{,}001\,m^3/kg$ $\quad v_2 = 0{,}0264\,m^3/kg$

$r = 0.15\,m$ $\quad P_{toplinska} = ? \rightarrow J/s \Rightarrow \dot{Q} = ?$

$\rho_1 A_1 c_1 = \rho_2 A_2 c_2$

(površina ista)

$$q_{12} + h_1 + \frac{c_1^2}{2} = h_2 + \frac{c_2^2}{2}$$

$\frac{1}{\rho} = v$

$\rho_1 \cdot c_1 = \rho_2 \cdot c_2$ $\quad c_1 = \frac{\rho_2 \cdot c_2}{\rho_1}$; $c_1 = \frac{\dot{m}_v}{\rho_1 \cdot A} = \frac{500\,kg/s \cdot 0{,}001}{(0{,}15)^2 \pi}$

$$c_2 = \frac{\dot{m}_p}{\rho_2 \cdot A} = \frac{500 \cdot 0{,}0264}{(0.15)^2 \pi}$$

$\dot{m}_p = \dot{m}_v$

$\boxed{c_1 = 7{,}07\,m/s}$

$\boxed{c_2 = 186.74\,m/s}$

$\dot{Q} = \left[\frac{J}{s}\right]$ $\quad q = \left[\frac{J}{kg}\right]$ $\rightarrow \dot{Q} = q \cdot \dot{m}$

$$q_{12} = h_2 - h_1 + \frac{1}{2}(c_2^2 - c_1^2) = 3100 - 134.7 + \frac{1}{2}(186.74^2 - 7{,}07^2)$$

$q_{12} = 2.98\,MJ/kg \xrightarrow{\cdot \dot{m}} \boxed{\dot{Q} = 1491.36\,MJ/s}$

Zadatak 7.)

7. U kondenzatoru se termoelektrane 3000 MJ/s topline predaje rashladnoj vodi. Temperatura je zasićene smjese vode i pare 30 °C, a temperatura rashladne vode 20 °C. Koliko se eksergije svake sekunde u kondenzatoru odvodi u okolicu? Koliki je prirast entropije okolice?

8. Na ulazu u sapnicu statora parne turbine, tlak je vodene pare 3 MPa, temperatura 500 °C, a brzina 30 m/s. Na izlazu je iz sapnice brzina 200 m/s, a tlak 100 kPa. Površina je ulaza u sapnicu 80 cm2 . Odredite maseni protok pare kroz sapnicu, temperaturu pare na izlazu iz sapnice i izlaznu površinu sapnice. Paru smatrajte idealnim plinom, R = 0.287 kJ/kgK, cp = 1.02 kJ/kgK, a strujanje jednodimenzionalnim i stacionarnim procesom. Razliku potencijalne energije pare na izlazu i ulazu u sapnicu zanemarite, kao i odvođenje toplinske energije u okolicu za vrijeme strujanje pare kroz sapnicu.